# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures for questions on heights and distances. Take $\sqrt{3} = 1.732$ where needed.

### SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** $\tan 30^\circ \times \tan 60^\circ$ equals: 1

**(a)** 1 **(b)** $\sqrt{3}$ **(c)** 0 **(d)** $\frac{1}{\sqrt{3}}$

**2.** If $\cos A = \frac{3}{5}$, then $\sec A$ is: 1

**(a)** $\frac{4}{5}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{5}{4}$

**3.** $\sin^2 A + \frac{1}{1+\tan^2 A}$ equals: 1

**(a)** 0 **(b)** $2\sin^2 A$ **(c)** $\sec^2 A$ **(d)** 1

**4.** A student states: "$\sin 60^\circ = 2\sin 30^\circ$." The correct response is: 1

**(a)** Correct for all angles **(b)** Correct

**(c)** Incorrect — $\frac{\sqrt{3}}{2} \neq 1$ **(d)** Cannot be determined

**5.** For $\csc 30^\circ$, Student A says 2; Student B says $\frac{1}{2}$. Then: 1

**(a)** Student A is correct **(b)** Student B is correct

**(c)** Both are correct **(d)** Both are wrong

**6.** A 15 m pole casts a shadow $5\sqrt{3}$ m long. The sun's elevation is: 1

**(a)** $90^\circ$ **(b)** $45^\circ$ **(c)** $30^\circ$ **(d)** $60^\circ$

**7.** A 15 m ladder makes $60^\circ$ with the wall. The height it reaches on the wall is: 1

**(a)** $7.5\sqrt{3}$ m **(b)** 15 m **(c)** 7.5 m **(d)** $5\sqrt{3}$ m

**8.** The angle of depression of a boat from a 50 m cliff is $45^\circ$. Its distance from the cliff is: 1

**(a)** 50 m **(b)** $\frac{50}{\sqrt{3}}$ m **(c)** 25 m **(d)** $50\sqrt{3}$ m

**9.** A student states: "If the shadow doubles, the sun's elevation is halved." The correct response is: 1

**(a)** Correct **(b)** Incorrect — $\tan\theta$ halves, not $\theta$

**(c)** Correct for $45^\circ$ **(d)** Cannot be determined

**10.** A tower of height $h$ is seen at $60^\circ$ from a distance $d$. Student A writes $h = d\sqrt{3}$; Student B writes $h = \frac{d}{\sqrt{3}}$. Then: 1

**(a)** Student B is correct **(b)** Student A is correct

**(c)** Both are correct **(d)** Both are wrong

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** If $5\tan\theta = 4$, find $\frac{5\sin\theta - 3\cos\theta}{5\sin\theta + 2\cos\theta}$. **2**

**12.** If $\sqrt{3}\tan\theta = 1$, find $\sin^2\theta - \cos^2\theta$. **2**

**13.** The angle of elevation of the top of a tower from a point 30 m away is $60^\circ$. Find its height. **2**

**14.** From the top of a 20 m building, the angle of depression of a car is $30^\circ$. Find the distance of the car from the building. **2**

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Prove that $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \sin A + \cos A$. **3**

**16.** The angles of elevation of the top of a tower from two points 30 m apart on the same side of it, in line with its foot, are $30^\circ$ and $60^\circ$. Find the height of the tower. **3**

**17.** If $\sec\theta = x + \frac{1}{4x}$, prove that $\sec\theta + \tan\theta = 2x$ or $\frac{1}{2x}$. **3**

**OR**

Prove that $\tan^2\theta - \sin^2\theta = \tan^2\theta\sin^2\theta$. **3**

**18.** A girl flies a kite with a 150 m string at $30^\circ$ to the ground; a boy flies one with a 100 m string at $45^\circ$. Whose kite is higher and by how much? **3**

**OR**

From the top of a 60 m lighthouse, the angles of depression of two boats on opposite sides of it are $30^\circ$ and $45^\circ$. Find the distance between the boats. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Prove that $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\csc A - 1}{\csc A + 1}$. **5**

**OR**

Prove that $\left(1 + \frac{1}{\tan^2 A}\right)\left(1 + \frac{1}{\cot^2 A}\right) = \frac{1}{\sin^2 A - \sin^4 A}$. **5**

**20.** **Case study:** A hot-air balloon rises vertically from a point A. From a point B, 200 m from A on level ground, its angle of elevation changes from $30^\circ$ to $60^\circ$ in 2 minutes. **5**

(i) Find the initial height.

(ii) Find the final height.

(iii) Find the height gained.

(iv) Find the speed of ascent.

**OR**

A man at the top of a 100 m tower sees a car moving towards the tower. The angle of depression changes from $30^\circ$ to $45^\circ$ in 10 seconds. Find the distance travelled and the speed of the car. **5**

## END OF QUESTION PAPER — Total: 40 Marks